

Piotr Brozek

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EXPERIENCE

Software Engineer Intern, AI/ML Platform

May 2026 – Aug. 2026

Tempus AI

Chicago, IL

- Benchmarked 5+ AI agent frameworks (LangChain, Claude SDK, Google ADK, CrewAI) across 12 dimensions to define enterprise standards for tool routing and SSE streaming; presented to 200+ engineers and senior leadership.
- Architected and deployed `deep_agent_sdk`, a production Python package wrapping LangChain/LangGraph with subagent tracing, semantic tool retrieval, SQLite memory, and human-in-the-loop interrupts; integrated into a centralized benchmark monorepo with live tool integrations (Slack, Jira, Confluence) via Okta, LiteLLM, and MCP brokers, and adopted as the proof-of-concept standard in production.
- Engineered production-grade LLM observability with OpenTelemetry and OpenInference, routing distributed traces to Arize Phoenix to monitor tool invocations, token usage, and guardrail spans.
- Reduced tool-invocation latency by 80% via caching strategies including MCP server pre-warming and cached LangChain node execution.

Research Assistant / Software Engineer

June 2024 – May 2025

Electronic Visualization Laboratory

Chicago, IL

- Enhanced SAGE3, a collaborative whiteboard used by 5,000+ users, in an Agile/Scrum team using TypeScript, React, and Docker-based cloud deployment.
- Built a LangChain-based LLM pipeline integrating DeepPlot visualizations, speaker diarization, and Llama 3 summarization to improve transcription accuracy and reduce manual summarization by 50%.
- Developed AI-powered export tools with Python and React to generate summaries as CSVs, PDFs, images, and sticky notes.

Full-Stack Developer Intern

Oct. 2024 – Feb. 2025

Ingenio Care Inc

Lombard, IL

- Developed a cross-platform physician-finder tool with React, Redux, Node.js, and Material UI for locating providers and booking appointments.
- Built an AI-powered insurance authorization system using OpenAI API, handling complex database management in PostgreSQL to reduce processing time by 30%.
- Contributed to frontend development of new features using Material UI, enhancing user experience and interface responsiveness across the platform.

PROJECTS

MRI Brain Tumor Detection and Classification | *Python, TensorFlow, Keras, NumPy*

- Built a Python ML model classifying brain tumors across 10,000+ MRI scans, combining ResNet-18 feature extraction with AdaBoost to boost accuracy on small clinical datasets.
- Preprocessed data with Pandas, applied transfer learning, and trained models on GPU with cross-validation, hyperparameter tuning, and performance visualized via Matplotlib.
- Demonstrated that the hybrid approach significantly improved classification accuracy in limited-data scenarios compared to standalone CNNs.

Clinical Trials Matching | *Python, FastAPI, LangGraph, LangChain, Ollama, React, Next.js, SQLite*

- Built a full-stack trial-matching platform with FastAPI and LangGraph, pairing a deterministic rules engine with local LLM (Ollama) review to screen 108 patients against 7,964 recruiting trials.
- Engineered a SHA-256 prompt-keyed cache and optimized SQLite indexing/batching, cutting fully-reviewed match latency 99.98% (140s to 25ms) and cohort rebuild time 76%.
- Developed a React/Next.js frontend with CopilotKit over the AG-UI protocol, streaming live agent tool calls into a chat interface, data explorer, and interactive screening board.

TECHNICAL SKILLS

Languages & Databases: JavaScript, TypeScript, Python, SQL, PostgreSQL, SQLite

Frameworks & Stacks: React, Node.js, Next.js, TensorFlow, LangChain, LangGraph, CrewAI

Developer Tools & Concepts: Cloud Deployment (Docker), Model Context Protocol (MCP), OpenTelemetry, Software Architecture Design, Agile/Scrum

CERTIFICATIONS

IBM Agentic AI, RAG & MCP Certification Series — IBM, May 2026

AI in Healthcare — Stanford University, June 2026

EDUCATION

University of Illinois at Chicago

May 2026

Bachelor of Science in Computer Science; GPA: 3.87/4.0; Magna Cum Laude

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